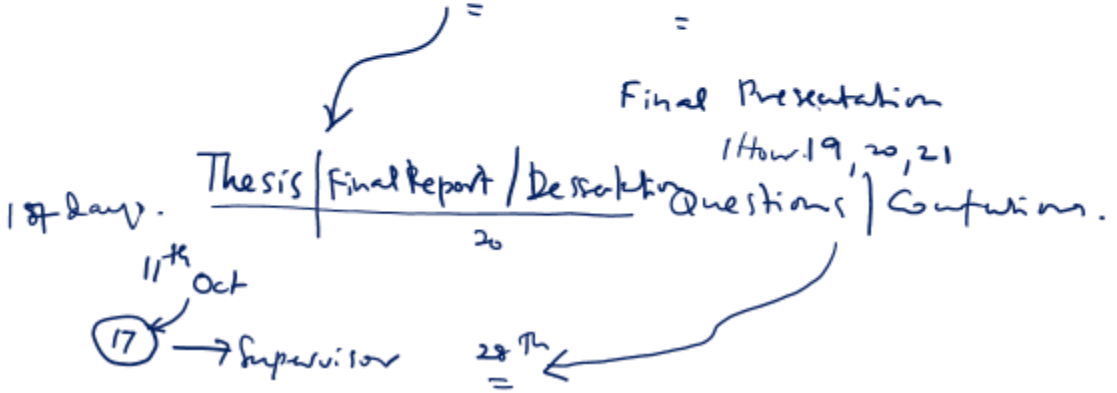
Final Product Completion, Testing & User Acceptance

IT4010 Research Project | Semester 2 Lecture 7

After PP2: The Mindset Shift

PP2	Final Assessment
Complete or near-complete system	→ Stable final product
Initial research evidence	→ Completed research evidence
Known weaknesses	User validation
Evaluator feedback received	Defensible thesis and viva



Do not demonstrate the same maturity level again.

What Defines a Final Product?





All promised core functions work end-to-end.

Components operate as one coherent solution.

Important non-functional requirements are demonstrated.

The intended users can perform realistic tasks.

Solution - Real World Problem
=
↓
usr/people

Functional Completion: Show One Complete Journey



Do not demonstrate four isolated modules.

Remove hard-coded outputs, manual file copying and mock responses.

Ask: can an unfamiliar user complete the main workflow without your help?

Integration Maturity

Weak Integration	Strong Integration
Separate logins	Shared data model
Duplicate databases	Clear APIs/interfaces
Manual data transfer	End-to-end workflow
Different UI styles	Consistent error handling
Components fail silently	One solution story

Testing: From “It Works” to “We Tested It”



Each level answers a different question.

Evidence may include test cases, logs, screenshots, bug fixes and supervisor/user feedback.

Testing should support the final thesis results and discussion.

Non-Functional Requirements: Measure Quality

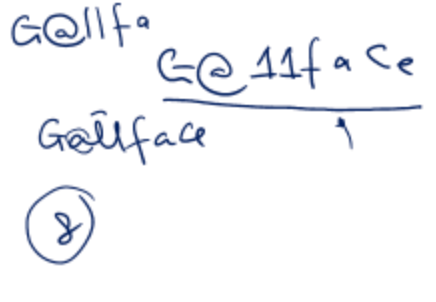
Performance: response time, throughput, inference time

Reliability: error handling, recovery, failure rate

Security: authentication, authorization, data protection

Usability: task success, user errors, SUS (System Usability Scale) where appropriate

Maintainability: modular code, documentation, deployment clarity



Do not list NFRs only. Demonstrate evidence.

Evaluation vs Validation

Evaluation

How well does it perform?

Accuracy / F1 / response time

SUS / task time / error rate

Baseline comparison where appropriate

Validation

Does it solve the intended problem?

Realistic user context

Stakeholder feedback

Evidence that solution is useful

Evaluation produces measurements. Validation asks whether those measurements support the intended solution.

User Acceptance: Let Users Perform Tasks

Do not only ask: “Do you like it?”

Prepare realistic tasks and expected outcomes.

Observe task completion, time, errors and confusion.

Record comments, issues and improvement actions.

Use this evidence in the thesis and final viva.

User acceptance is stronger when users perform real tasks.

Evidence Needed for Final Presentation & Viva

- ✓ Working integrated system
- ✓ Test results
- ✓ User feedback / UAT
- ✓ Research evaluation
- ✓ GitHub history
- ✓ Planner progress
- ✓ RP diary
- ✓ Supervisor feedback
- ✓ AI-use record
- ✓ Thesis draft evidence

Supervisor Meetings + Teams = Traceable Progress



Bring updated build, test results, thesis sections and blockers to each meeting.

Use MS Teams for meeting records and decisions.

Use Planner for tasks and owners; use GitHub for technical history.

Use the RP diary to record individual decisions and learning.

AI Use and Academic Integrity

Acceptable Support

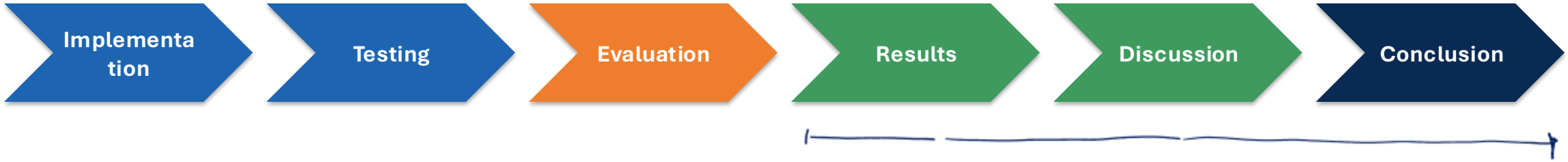
- Debugging ideas
- Test-case suggestions
- Language improvement
- Concept explanation
- Documentation support

Student Responsibility

- Verify outputs
- Disclose usage
- Understand generated code
- Check citations
- Defend decisions

AI may assist your work. It cannot become the owner of your work.

How Today's Evidence Feeds the Thesis



Testing evidence supports implementation and quality claims.

Evaluation results support research objectives and findings.

User acceptance supports validation and real-world relevance.

Limitations and failures strengthen the discussion when explained honestly.

↳ Future Directions
Works

Final 5-Minute Readiness Audit

Score your team from 1–5 for: functionality, integration, stability, testing, NFR evidence, user validation, research evidence, supervisor review, documentation and individual ownership.

Identify the weakest three areas.

Assign owner, deadline and evidence for each improvement.

Closing message: PP2 showed progress. Final must prove completion.